

ARK SCIENTIFIC			
INSTALLATION QUALIFICATION			
Title	Fluidized Bed Drier	Model No	ARK FBD-Lab
Protocol Number	ARK /IQ/01	Page No.	1 of 15
Effective Date		Sr No	

INSTALLATION QUALIFICATION PROTOCOL Fluidized Bed Drier

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1.0 PRE-APPROVAL :

Signing of this Approval page of Installation Qualification Protocol No. ARK /IQ/01, indicates agreement with the Installation Qualification approach described in this document. If Modifications to the Installation Qualification become necessary, an addendum will be prepared and approved.

Name	Signature	Date
Written By: ARK SCIENTIFIC		
Checked By:		
Approved By:		

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2.0 OVERVIEW

Purpose:

- ❖ To provide record of the key feature of the equipment and components as currently installed.
- ❖ To ensure that there is sufficient information available to enable the equipment to be operated and maintained safely, effectively and consistently.
- ❖ To verify the Installation attributes of the Fluidized Bed Drier (supplied by ARK Scientific) critical to serve the intended purpose for the equipment.
- ❖ To establish the suitability of the draft SOP prepared for the operation of equipment.
- ❖ To document the observation for future reference.

2.1 SCOPE:

The scope of this qualification protocol is limited to the installation qualification Fluidized Bed Drier. This qualification protocol is part of a validation activity.

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2.2 RESPONSIBILITY :

The validation group comprising of :

Sr.No	Name of Person	Designation
1)		
2)		
3)		

2.3 REQUALIFICATION :

Installation Qualification to be repeated in case of:

- ◆ Replacement of any major component / instrument.
- ◆ Major modification in the existing equipment.
- ◆ During monitoring, if equipment is found to be malfunctioning.
- ◆ Shifting of the equipment from one location to another.
- ◆ As per Requalification Schedule.

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3.0 **PROCEDURE:**

After unpacking, the machine should be carefully inspected & notified immediately of any suspected damage to the machine (if any). The machine should then be checked for loose fasteners and if any should be placed in its desired location. The machine should be leveled properly. The machine should be cleaned thoroughly. All necessary adjustments and settings have already been done at our factory for convenient installation. Only minor re-adjustments may be required to suit particular conditions and requirements. Check the utility requirements as per specifications and connect the machine to the recommended power.

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3.1 PHYSICAL VERIFICATION:

Sr. No	Description	Specification as per URS	Observations	Checked by/Date
1.	Make	ARK Scientific		
2.	Model	ARK FBD-Lab		
3.	Container Volume	6 Litre – SS Bowl Glass Bowl – 6 Lit 0.3 x 3 lit Glass Bowl		
4.	Material of Construction	Contact parts- SS316 & Non-contact part- SS304		
5.	Electrical construction	Non flame proof		
6.	Desirable material charging/loading method into equipment	Manual		
7.	Desirable material discharging/unloading method into equipment/ system	Manual		
8.	Types of Control	Manual		
9.	Parameters settable from the control panel	Temperature, time & Air flow		
10.	Parameters to be displayed on the control panel	Temperature, time & Air flow		

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Sr. No	Description	Specification as per URS	Observations	Checked By Sign/Date
11.	Electrical Supply	1 Phase – 230 Volts, 50 Hz		
12.	Safety Control	Provision of overheat control		
		Provision of MCB to be provided so that when there is an overload in current or any short circuit then the MCB Trips.		
13.	Overall Dimension	460mmX560mmX 578mm (approx.)		
14.	Weight	Approximate 25 Kg		
15.	Filter Bag	Make- SEFAR		
16.	HMI	Make- Delta		

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3.2 DOCUMENT VERIFICATION:

Sr. No.	Description	Test Certificate No.	Check By Sign/Date
1	GA drawing	DFM-000	
2	Electrical Circuit Drawing	LE-20/159	
3	MOC certificates	-	

4.0 PLC DESCRIPTION:

The main function of a PLC is to translate the instruction into the digital or analog codes needed to operate the device or machine.

The main function of this PLC system is to collect data from field instrumentation & display the information on the operator station. The instruments are connected to the system equipment and piping. The collected data will be utilized by the PLC for process control.

The user interface operating panel, based HMI (Human - Machine Interface), will assist the operator to supervise and control the process. Based on the displayed information the operator, by means of the user interface, can provide commands to the PLC.

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4.1 Digital Input from PLC (DVP14SS211R)

X0	Emergency
X1	Spare
X2	Spare
X3	Spare

4.2 Digital Outputs from PLC (DVP14SS211R)

Y0	Air Flow
Y1	Heater Safety
Y2	Buzzer
Y3	Spare

4.3 Analog Card (DVP04DA-S)

01	Analog o/p of Air Flow to Speed Control
02	Spare
03	Spare
04	Spare

4.4 DTC Output

Out 1	Status
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5.0 ACTION PLAN IN CASE OF DEVIATIONS:

If there is no deviation, then write N.A.

Description of deviation and date observed:

Person, responsible for corrective action and date assigned:

Corrective actions taken and date conducted:

6.0 ACCEPTANCE CRITERIA:

Operational Qualification shall be considered acceptable when all the conditions specified under section 3.0 have been met.

Any deviation from the acceptance criteria of the specific check point shall be reported and decision should be taken for the rejection, replacement or rectification of the instrument/component.

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9.0 REVISION HISTORY:

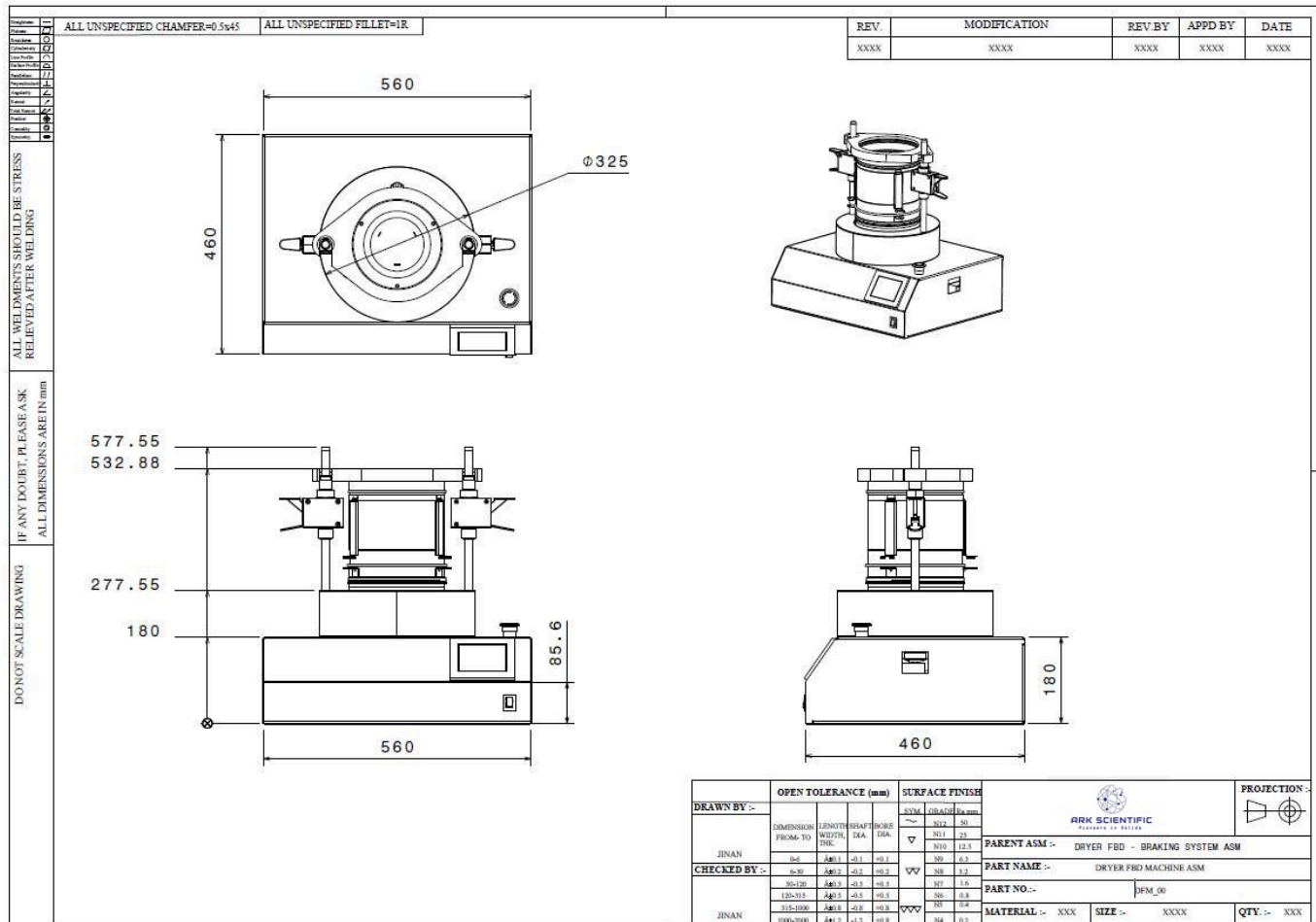
Sr.No.	Revision No.	Revision Detail	Change Control No.
1	00		

10.0 POST APPROVAL:

Name	Designation	Department	Signature	Date

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11.0 G.I. Drawing



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12.0 CERTIFICATION:

CERTIFICATE

This is to certify that Installation Qualification of

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Machine ID. No:

Has been completed and it confirms to all the

Acceptance criteria of IQ.

Hence, released for Operational Qualification purpose.

(Sign. / Date)	(Sign. / Date)	(Sign. / Date)
Certified By (Engineering)	Certified By (Production)	Approved By (QA)